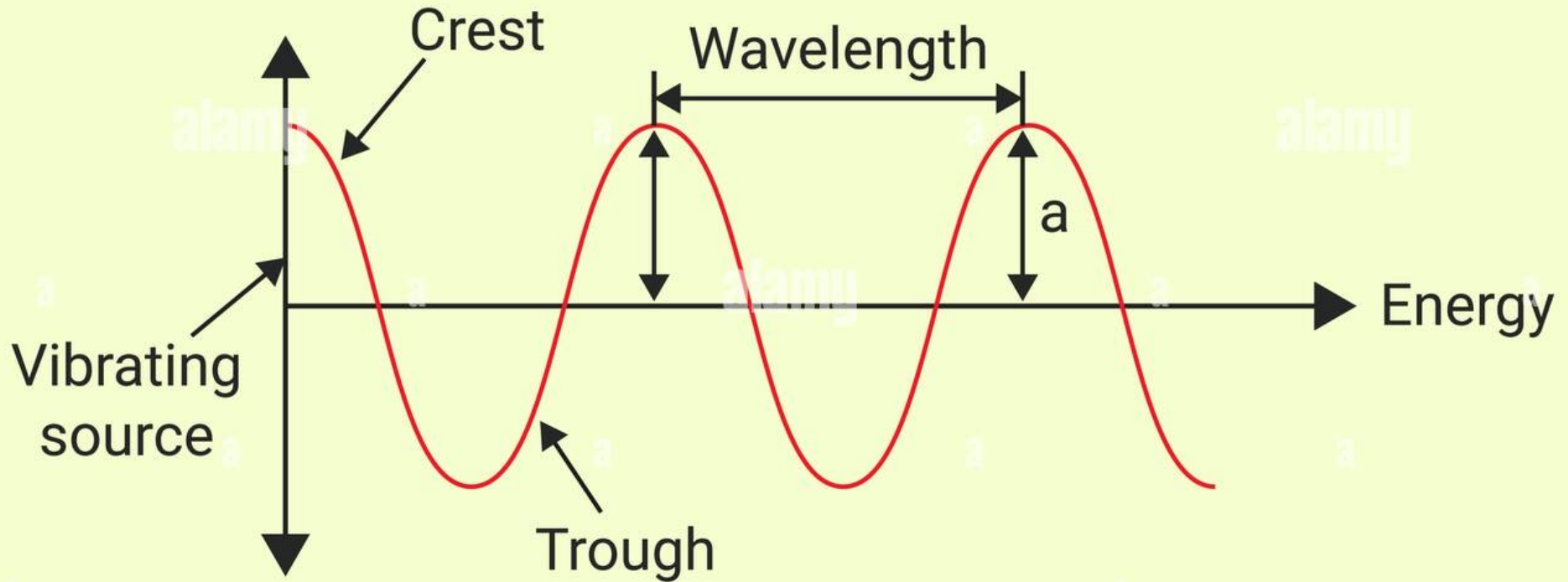


**What do these have in common?**



# What is a Wave?

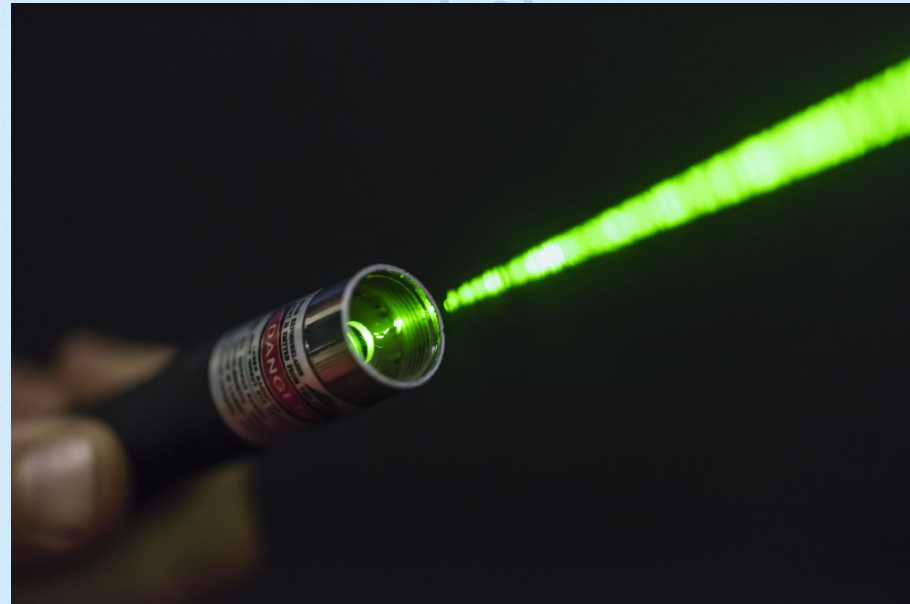
A disturbance that transfers energy without transferring matter.  
Waves carry information and/or energy from place to place.



# Two Big Families of Waves

Mechanical: need some medium (air, water, rope). Examples: sound and ocean waves.

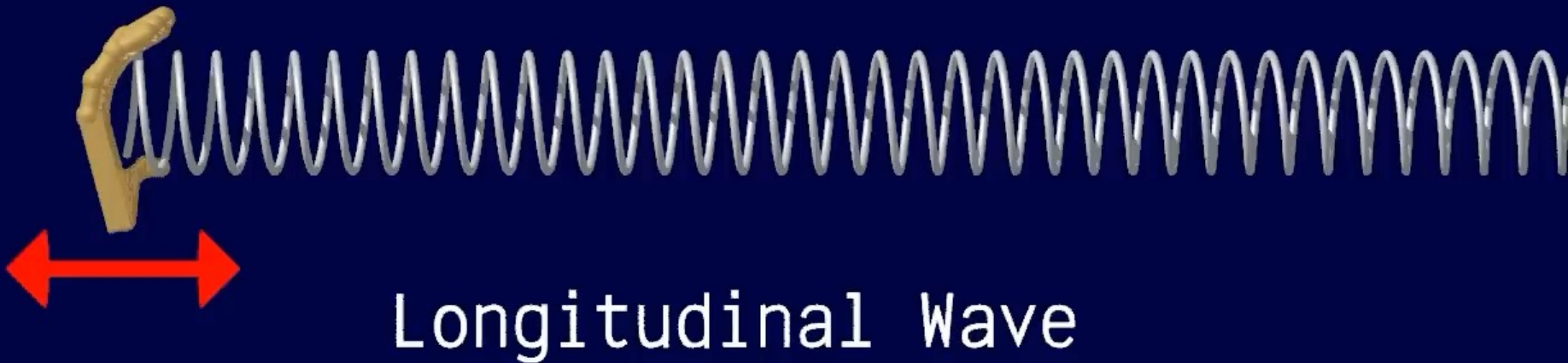
Electromagnetic (EM): no medium needed; can travel in vacuum. Examples: light and radio.



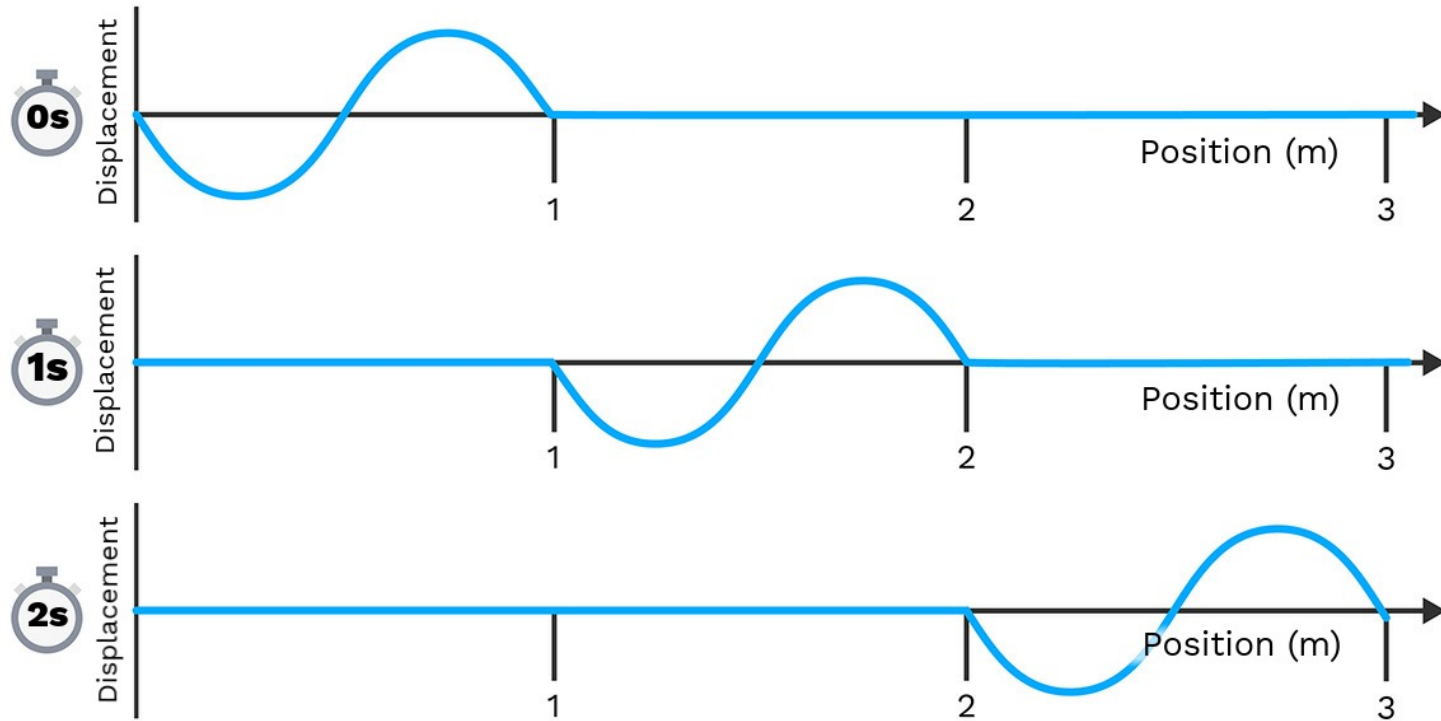
# Wave Shapes: Transverse vs Longitudinal

Transverse: particles move up/down (or sideways) while the wave moves forward (light, water surface).

Longitudinal: particles compress/expand along wave direction (sound in air).



# Speed Link

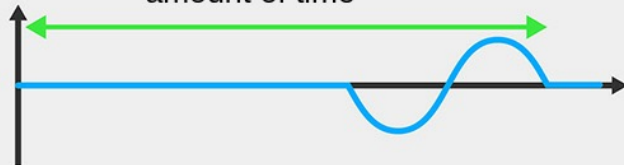


The speed of this wave is 1m/s



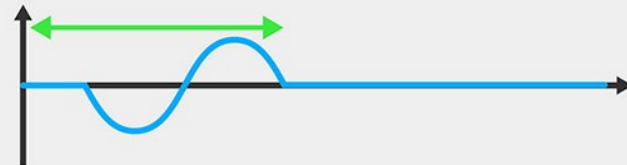
## Faster speed

More distance in the same amount of time



## Slower speed

Less distance in the same amount of time



Sort the waves!

Mechanical/Electromagnetic? Transverse/Longitudinal?



**Sort the waves!**

~~Visible Light~~/Electromagnetic? Transverse/~~Longitudinal~~?



2

Sort the waves!

Mechanical/Electromagnetic? Transverse/Longitudinal?



Sort the waves!

Mechanical/Electromagnetic? Transverse/Longitudinal?



Sort the waves!

Mechanical/Electromagnetic? Transverse/Longitudinal?



Sort the waves!

~~Mechanical~~/Electromagnetic? ~~Transverse~~/Longitudinal?



Sort the waves!

~~Mechanical~~/Electromagnetic? Transverse/~~Longitudinal~~?



Sort the waves!

~~Microwave~~/Electromagnetic? Transverse/Longitudinal? ~~Longitudinal~~

